

Quiz (Carolina Reaper)

- Q1.** What is the purpose of a two-way frequency table?
- (A) To display one categorical variable
(B) To display two numerical variables
(C) To analyze the relationship between two categorical variables
(D) To calculate the mean of a dataset
(E) To create a bar graph
- Q2.** Given a two-way table with 3 rows and 4 columns, how many cells are there in total?
- (A) 6
(B) 10
(C) 12
(D) $3 \times 4 = 12$
(E) 16
- Q3.** Which of the following is a type of relative frequency?
- (A) Joint relative frequency
(B) Marginal relative frequency
(C) Conditional relative frequency
(D) All of the above
(E) None of the above
- Q4.** What is the formula for joint relative frequency?
- (A) $\frac{a}{n}$
(B) $\frac{a}{r}$
(C) $\frac{a}{c}$
(D) $\frac{a}{n}$, where a is the frequency and n is the total
(E) $\frac{r}{c}$
- Q5.** In a two-way table, what does the marginal relative frequency represent?
- (A) The proportion of data in each cell
(B) The proportion of data in each row
(C) The proportion of data in each column
(D) The proportion of data in each row and column
(E) The proportion of data in the entire table
- Q6.** What is the purpose of calculating conditional relative frequencies?
- (A) To analyze the relationship between two categorical variables
(B) To compare the proportions of data in each row or column
(C) To calculate the joint relative frequency
(D) To determine the marginal relative frequency
(E) To create a bar graph
- Q7.** Given the following two-way table:

	A	B	Total
X	10	20	30
Y	15	25	40
Total	25	45	70

 What is the joint relative frequency of A and X?
- (A) $\frac{10}{70}$
(B) $\frac{10}{30}$
(C) $\frac{10}{25}$
(D) $\frac{10}{10}$
(E) $\frac{25}{70}$
- Q8.** Which of the following statements is true about the relationship between two categorical variables?
- (A) If the variables are independent, then the conditional relative frequencies will be equal
(B) If the variables are dependent, then the joint relative frequencies will be equal
(C) If the variables are independent, then the joint relative frequencies will be equal
(D) If the variables are dependent, then the marginal relative frequencies will be equal
(E) None of the above

Open Ended Questions (FRQ)

- Q9.** Suppose we have the following two-way table:

	A	B	Total
X	12	18	30
Y	20	30	50
Total	32	48	80

 Calculate the joint relative frequency of A and X, and the conditional relative frequency of A given X.
- Q10.** Using the same table as in question 9, calculate the marginal relative frequencies for the rows and columns, and explain what they represent.

Chef's Tips

- Make sure students have a calculator and pencil
- Allow 20 minutes to complete the quiz
- Remind students to show all work for free response questions